Srujal Jitendrakumar Patel

Data Scientist (AI/ML)

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SUMMARY

Data Scientist (AI/ML) with 3+ years of experience designing and deploying machine learning solutions, predictive analytics models, and enterprise AI applications across healthcare and business domains. Experienced in developing scalable data pipelines, feature engineering workflows, statistical modeling, deep learning, natural language processing, Generative AI, Retrieval-Augmented Generation (RAG), and LLM-powered applications using Python, SQL, Spark, TensorFlow, PyTorch, and cloud platforms. Skilled in transforming structured and unstructured data into actionable insights, optimizing model performance, implementing MLOps best practices, and collaborating with cross-functional teams to deliver production-ready AI solutions that improve operational efficiency, business intelligence, and customer outcomes.

PROFESSIONAL EXPERIENCE

Data Scientist (AI/ML) | Cigna | USA — Aug 2025 – Present

- Designed and developed machine learning models using Python, Scikit-learn, XGBoost, and TensorFlow to predict healthcare utilization, patient risk scores, and claims outcomes across datasets containing over 6 million records, improving prediction accuracy by 24% and supporting data-driven clinical decisions.
- Built scalable data preprocessing and feature engineering pipelines using PySpark, SQL, Databricks, and Pandas to cleanse and integrate healthcare, claims, provider, and member data, reducing data preparation time by 40% and improving model training efficiency.
- Developed Retrieval-Augmented Generation (RAG) solutions using LangChain, ChromaDB, and embedding models to enable secure AI-powered knowledge retrieval, improving contextual response relevance by 30% while reducing manual document search efforts.
- Collaborated with senior data scientists to fine tune transformer-based language models and implement prompt engineering techniques for healthcare AI applications, improving LLM response consistency by 22% while reducing hallucinations through structured validation.
- Automated machine learning workflows using MLflow, Docker, GitHub Actions, and CI/CD pipelines for model versioning, deployment, and monitoring, reducing model release cycles by 35% while improving deployment reliability.
- Performed exploratory data analysis and statistical modeling using Python, SQL, Pandas, and visualization tools to identify healthcare trends, patient segments, and cost drivers, delivering actionable insights for business and clinical teams.
- Collaborated with data engineers, software developers, and product teams to deploy REST APIs using FastAPI and integrate predictive models into enterprise applications, improving inference response time by 28% and supporting scalable AI solutions.
- Implemented Explainable AI techniques using SHAP and model validation frameworks to improve prediction transparency, monitor model performance, and support healthcare governance requirements, increasing stakeholder confidence in production AI models.

Data Scientist | Cognizant | India — July 2021 – Nov 2023

- Designed and deployed machine learning models using Python, Scikit-learn, XGBoost, TensorFlow, and SQL to solve customer analytics, fraud detection, and business forecasting use cases, improving model accuracy by 27% while supporting data-driven business decisions.
- Built end-to-end ETL and data engineering pipelines using PySpark, Apache Spark, Airflow, SQL, and Databricks to process structured and unstructured data from more than 30 enterprise sources, reducing data processing time by 42% and improving data availability.
- Performed extensive feature engineering, exploratory data analysis, and statistical modeling using Pandas, NumPy, and Python to identify key business drivers, improving predictive model performance by 25% while increasing data quality across multiple AI initiatives.
- Developed deep learning and Natural Language Processing (NLP) models using TensorFlow, PyTorch, Transformers, and BERT for text classification, sentiment analysis, and document intelligence solutions, increasing prediction accuracy by 23% and reducing manual review effort.
- Built and deployed scalable machine learning models using MLflow, Docker, FastAPI, and REST APIs, enabling automated model serving and monitoring while reducing deployment time by 35% and improving production stability.
- Worked with data engineers, business analysts, and product teams to develop interactive dashboards using Power BI and Tableau, translating machine learning insights into executive reports that reduced reporting effort by 40% and supported strategic planning.
- Optimized cloud-based AI workflows using AWS services, Snowflake, Git, and CI/CD pipelines to automate model training, validation, and deployment, improving development efficiency by 30% while ensuring scalable and reliable production releases.
- Applied model monitoring, Explainable AI (SHAP), A/B testing, and performance validation techniques to evaluate production models, improve prediction transparency, and maintain high model reliability, reducing model drift by 20% across deployed machine learning solutions.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, R, Scala, Bash.
- **Machine Learning:** Supervised Learning, Unsupervised Learning, Ensemble Learning, XGBoost, LightGBM, CatBoost, Scikit-learn.
- **Deep Learning & AI:** TensorFlow, PyTorch, Keras, CNN, RNN, LSTM, Transformers, Transfer Learning.
- **Generative AI & LLMs:** OpenAI API, Claude API, LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Prompt Engineering, Vector Databases.
- **Data Engineering:** Apache Spark, PySpark, Databricks, Airflow, Snowflake, Delta Lake, ETL Pipelines.
- **Databases:** PostgreSQL, MySQL, SQL Server, MongoDB, ChromaDB, Pinecone.
- **Cloud & MLOps:** AWS, Azure, Docker, Kubernetes, MLflow, GitHub Actions, CI/CD.
- **Data Analysis & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Plotly, Power BI, Tableau.
- **Statistical & Predictive Modeling:** Hypothesis Testing, Time Series Forecasting, Regression, Classification, Clustering, A/B Testing.
- **Healthcare Analytics:** Clinical Data Analysis, Claims Analytics, Risk Prediction, Population Health Analytics, HIPAA Compliance.
- **Development Tools:** Git, GitHub, Jupyter Notebook, VS Code, FastAPI, REST APIs, Linux.
- **Professional Skills:** Feature Engineering, Model Deployment, Explainable AI (SHAP), Model Monitoring, Cross-functional Collaboration, Agile, Scrum.

EDUCATION

Master's in Computer Engineering | Arizona State University, Tempe, AZ, USA — **Jan 2024 – Dec 2025**

Bachelor's in Electronics and Communication Engineering | L.D. College of Engineering, Gujarat, India — **Aug 2019 – Jun 2023**

PROJECTS

Enterprise Healthcare Risk Prediction Platform

- Developed an end-to-end machine learning solution using Python, Scikit-learn, XGBoost, PySpark, and SQL to predict patient readmission risk and healthcare utilization by analyzing clinical, claims, and demographic datasets, improving prediction accuracy by 26%.
- Built automated feature engineering, model training, and deployment pipelines using MLflow, Docker, FastAPI, and GitHub Actions while implementing SHAP-based Explainable AI, reducing deployment time by 35% and improving model transparency for clinical stakeholders.

Generative AI Clinical Knowledge Assistant

- Designed a Retrieval-Augmented Generation (RAG) application using LangChain, ChromaDB, embedding models, OpenAI APIs, and FastAPI to provide secure access to enterprise healthcare documentation, improving contextual response relevance by 31% while reducing manual knowledge search time.
- Implemented prompt engineering, semantic search, document chunking, and vector indexing techniques to improve retrieval quality, while deploying the application using Docker and cloud infrastructure to support scalable enterprise AI workloads.